COSC 4326 Network Theory
Homework 4

(Refer to the attached textbook chapter to finish this homework)

Please submit your finished homework (typed using computers) to Blackboard by 11:59:00 pm CST on April 3, 2026.

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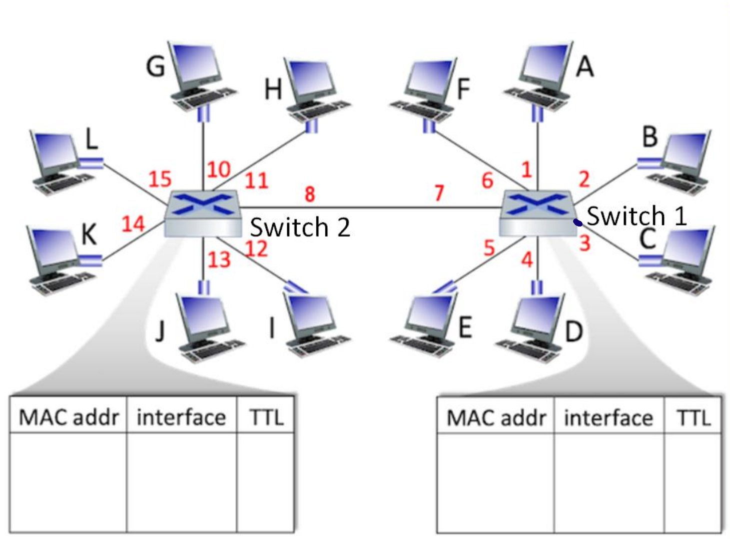
1. Consider the LAN below consisting of 10 computers connected by two self-learning Ethernet switches. At t=0 the switch table entries for both switches are empty. At t = 1, 2, 3, and 4, a source node sends to a destination node as shown below, and the destination replies immediately (well before the next time step).

   t=1: B -> D
   t=2: F -> E
   t=3: H -> D
   t=4: F -> C



1.1 Fill out the two switch tables.

Table of switch 2		
MAC addr	Interface	TTL
B	8	t=1 (low)
F	8	
H	11	
D	8	t=3 (high)

Table of switch 1		
MAC addr	Interface	TTL
B	2	t=1
D	4	
F	6	
E	5	t=4

1.2 Answer the questions below.

a) At t=1, what is the source entry for switch 1? Format your answer as: <letter, number>, 'n/a', or 'existed entry'.

<B, 2>

b) At t=1, what is the destination entry for switch 1? Format your answer as: <letter, number>, 'n/a', or 'existed entry'.

n/a

c) At t=1, what is the source entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. <B,8>

d) At t=1, what is the destination entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. n/a

e) At t=2, what is the source entry for switch 1? Format your answer as: <letter, number>, ‘n/a’, or 'existed entry'. <F,6>

f) At t=2, what is the destination entry for switch 1? Format your answer as: <letter, number>, ‘n/a’, or 'existed entry'. n/a

g) At t=2, what is the source entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. <F,8>

h) At t=2, what is the destination entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. n/a

i) At t=3, what is the source entry for switch 1? Format your answer as: <letter, number>, ‘n/a’, or 'existed entry'. <H,7>

j) At t=3, what is the destination entry for switch 1? Format your answer as: <letter, number>, ‘n/a’, or 'existed entry'. existed entry

k) At t=3, what is the source entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. <H,11>

l) At t=3, what is the destination entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. n/a

m) At t=4, what is the source entry for switch 1? Format your answer as: <letter, number>, ‘n/a’, or 'existed entry'. existed entry

n) At t=4, what is the destination entry for switch 1? Format your answer as: <letter, number>, ‘n/a’, or 'existed entry'. n/a

o) At t=4, what is the source entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. existed entry

p) At t=4, what is the destination entry for switch 2? Format your answer as <letter, number>, ‘n/a’, or 'existed entry'. n/a

Example:
t=1: C -> H
t=2: C -> J
t=3: C -> D
t=4: I -> J

a) At time t=1, (C,3) is added as an entry to switch table 1. t=1: C→H
b) At time t=1, (H,7) is added as an entry to switch table 1. t=1: C→H
c) At time t=1, (C,8) is added as an entry to switch table 2. t=1: B→D
d) At time t=1, (H,11) is added as an entry to switch table 2. t=1: B→D
e) At time t=2, since the entry for computer C in switch table 1 already exists, no new table entry is made t=2: F→E
f) At time t=2, (J,7) is added as an entry to switch table 1. t=2: F→E
g) At time t=2, since the entry for computer C in switch table 2 already exists, no new table entry is made t=2: F→E
h) At time t=2, (J,13) is added as an entry to switch table 2. t=2: F→E

i) At time t=3, since the entry for computer C in switch table 1 already exists, no new table entry is made t=3 : H→D
j) At time t=3, (D,4) is added as an entry to switch table 1. t=3 : H→D
k) At time t=3, since the entry for computer C in switch table 2 already exists, no new table entry is made t=3 : H→D
l) At time t=3, switch table 2 doesn't observe this frame (no one replies to switch 2) t=3 : H→D
m) At time t=4, switch table 1 doesn't observe this frame (switch 2 knows where is J, switch 2 does not forward frame to switch 1) t=4 F→C
n) At time t=4, switch table 1 doesn't observe this frame (switch 2 knows where is J, switch 2 does not forward frame to switch 1) t=4 F→C
o) At time t=4, (I,12) is added as an entry to switch table 2. t=4 F→C
p) At time t=4, since the entry for computer J in switch table 2 already exists, no new table entry is made t=4 : F→C

a) (C,3) : C's frame arrives at SW1 on port <u>3</u>.

b) (H,7) : H's reply crosses to SW1 on inter-switch port <u>7</u>

c) (C,8) : C's flooded frame arrives SW2 on port <u>8</u>

d) (H,11) : H's reply arrives SW2 on port <u>11</u>

e) no new entry : C=3 already learned at t=1.

f) (J,7) : J's reply crosses to SW1 on port <u>7</u>

g) no new entry : C=8 already learned at t=1.

h) (J,13) : J's reply arrives SW2 on port <u>13</u>.
to SW2

i) no new entry : C=3 already known

j) (D,4) : D's reply arrives arrives on port <u>4</u>

k) no new entry : C=8 already known

l) SW no reply : D's reply goes directly C via SW1 only never reach SW2

m) sw1 doesn't observe frame : sw2 knows J:13, never floods sw1

n) sw1 doesn't observe reply: everything stays in sw2.

o) (I,12) added to sw1 : I's frame arrives sw2 on port 12

p) no new entry for J in sw2: J=13 already learned at t=2.